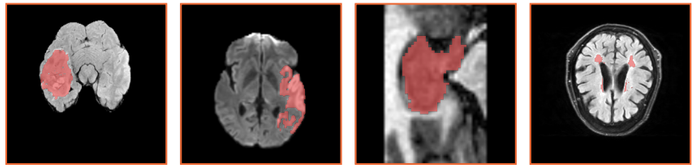
Which adaptation strategy for a new task? A rule that is knowable before training: does the task inherit a large head?

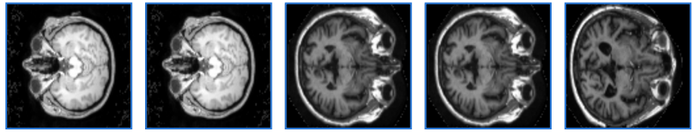
(a) Nine tasks, six cohorts, 32 labelled volumes each

Large task head: inherits the 8.45M-parameter decoder



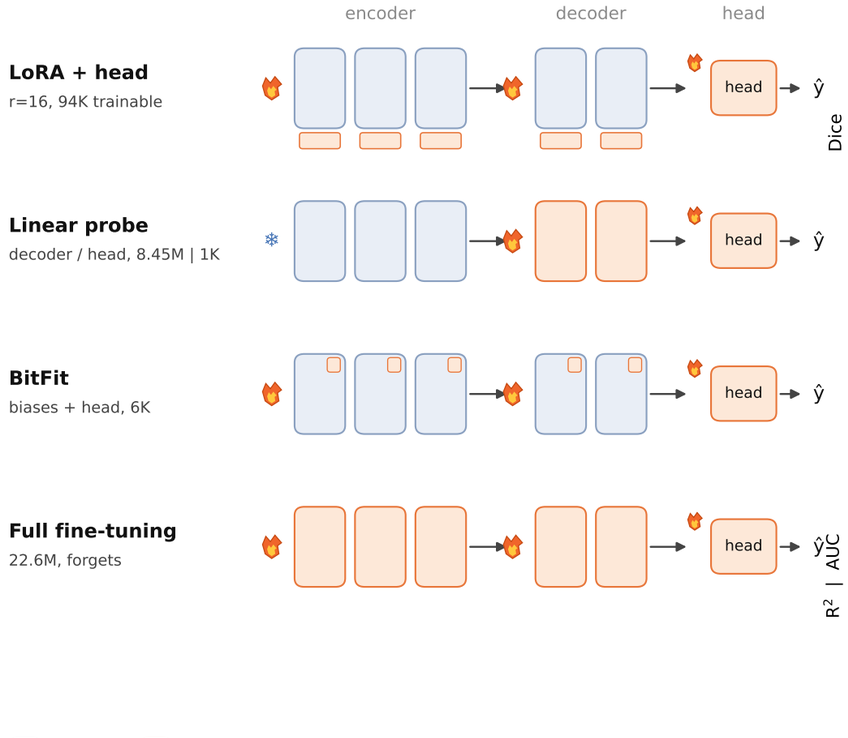
BraTS glioma | ISLES stroke | Hippocampus | WMH, 5 scanners

Small task head: ~1K-parameter head on pooled features

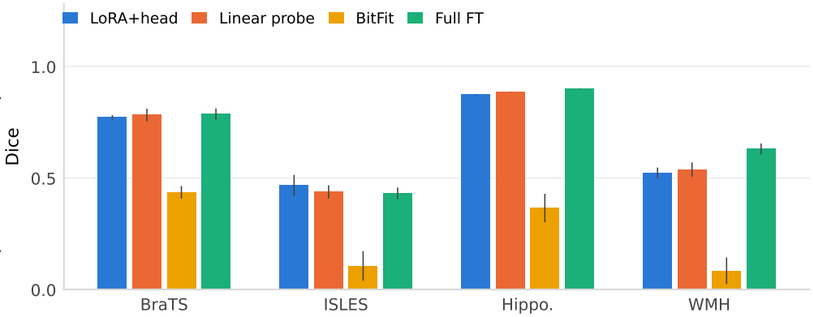


IXI age | IXI sex | OASIS age | OASIS sex | OASIS CDR

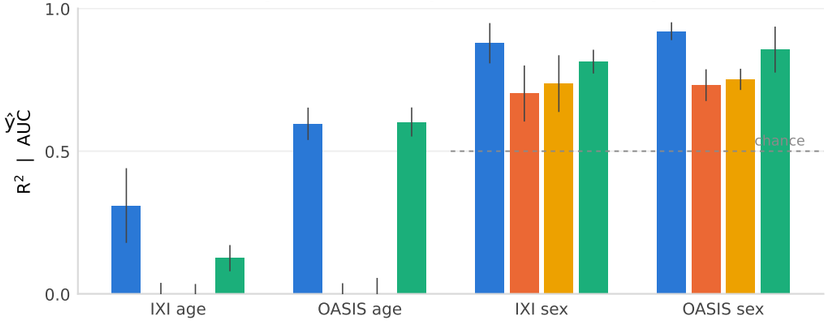
(b) Adaptation strategies: what is frozen and what is trained



(c) Large head → train it: LoRA within 0.015 Dice of the probe; BitFit collapses



Small head → only LoRA learns: probe and BitFit output a constant



Forgetting is symmetric: under full FT a second segmentation task erases the first in either order (BraTS 0.78→0.20, ISLES 0.40→0.01); EWC, LwF, replay do not prevent it; frozen strategies cannot forget.

Pretraining is task-dependent: vs. random init, ΔDice = +0.000 BraTS, -0.030 ISLES, +0.024 WMH; run a random-init control.

Protocol: subject-grouped splits, whole-volume inference, 5 seeds, 301 runs, paired per-case tests with Holm correction.